

VENKAT DINESH PASUPULETI

Applied ML Scientist · AI Engineer · LLM Applications · Document Intelligence · MLOps

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Windsor / Toronto, ON, Canada | Open Work Authorization until Nov 2026 | Immediate Start

6.5 Years | **97K** Lines Python | **90%** Fraud Precision | **50%** OCR Reduction | **Azure DP-100** Certified

PROFESSIONAL SUMMARY

- Applied ML Scientist and AI Engineer with 6.5 years of hands-on experience across the full ML lifecycle — from problem framing, data engineering, and model training through production deployment, monitoring, and iteration. Expertise spans Generative AI (LLMs, RAG, agentic systems), classical ML (XGBoost, Random Forest, CNNs), document intelligence (OCR, structured extraction), and production MLOps on Azure and GCP.
- Proven delivery across regulated domains: 4+ years in financial services (fraud detection 90% precision, loan approval, customer segmentation, OCR document automation), AI SaaS (97K lines Python, 557 FastAPI endpoints, production LLM voice AI serving real callers), telecom analytics, and agricultural ML (Azure ML migration, R-based statistical model operationalization).
- Strong engineering foundation: clean modular Python, FastAPI production APIs, GKE/Kubernetes autoscaling, GitHub Actions CI/CD, MLflow experiment tracking, Azure ML managed endpoints. DP-100 certified. Comfortable across the full stack from data ingestion to production endpoint with monitoring, drift detection, and retraining pipelines.
- Clear technical communicator — translate model trade-offs, performance metrics, and architecture decisions for both technical engineers and non-technical business/compliance stakeholders. Experience partnering with risk, compliance, audit, and operations teams in regulated environments (OSFI, PIPEDA, PHIPA-aware design).

TECHNICAL SKILLS

- **GenAI & LLMs:** OpenAI API (GPT-4.1, GPT-4o), Anthropic Claude (Sonnet 4.5), Google Gemini, Meta Llama 3.2, Mistral, LangChain, LangGraph, RAG Architectures, Agentic Workflows, Prompt Engineering, Tool-Calling / Function Calling, Structured JSON Extraction, Hallucination Prevention, Negative Example Anchoring, LLM Evaluation (RAGAS, LLM-as-judge)
- **ML & Deep Learning:** Scikit-learn, XGBoost, Random Forest, Extra Trees, SVM, TensorFlow, PyTorch, CNN (YOLOv3, Faster R-CNN, OpenCV), Transformers, Embeddings, Classification, Regression, Forecasting (ARIMA, SARIMA, Prophet), Clustering, Anomaly Detection (Isolation Forest), Recommendation Systems, Feature Engineering, SHAP, Hyperparameter Tuning, Cross-Validation
- **Document Intelligence & OCR:** Azure AI Document Intelligence (Form Recognizer), Azure ML OCR Pipelines, PDF Parsing (PyMuPDF), Table Extraction, Key-Value Extraction, Semi-Structured Document Processing, Financial Statement Parsing, Clinical Document Parsing, Identity Document Verification, Named Entity Extraction
- **Python Engineering:** Python 6.5 years (advanced), FastAPI, REST APIs, WebSocket, GraphQL, async/await, OOP, PEP Standards, Pydantic, SQLAlchemy, Clean Code, Modular Architecture, Service Registry Pattern, Pytest, Type Hints
- **R Programming:** R (statistical modeling), caret, randomForest, glmnet, xgboost (R), e1071, tidyverse, dplyr, ggplot2, data.table, Rmarkdown, time series (forecast package), regression, classification, R → Python model migration
- **Cloud — Azure:** Azure Machine Learning (Studio, Managed Endpoints, Compute Clusters, Model Registry, Pipelines), Azure DevOps CI/CD, Azure Blob Storage, Azure Data Factory, Azure AI Document Intelligence, Azure Monitor, Azure Application Insights, Azure Key Vault, AKS — Microsoft Certified DP-100
- **Cloud — GCP:** Vertex AI, Cloud Run, GKE (Kubernetes), BigQuery, Pub/Sub, Cloud Storage, IAM, Secret Manager, Cloud Functions — production SaaS deployment
- **MLOps & Deployment:** MLflow (experiment tracking, model registry, deployment), Docker, Kubernetes (GKE/AKS), GitHub Actions CI/CD, Azure DevOps, Model Versioning, Drift Detection (PSI, KS test), Prometheus, Grafana, p50/p95/p99 Monitoring, Cost per Inference, Automated Retraining Pipelines
- **Databases & Data Engineering:** PostgreSQL, MongoDB, Redis, Elasticsearch, SQL, Apache Spark, PySpark, Pandas, NumPy, ETL/ELT Pipelines, Data Quality Validation (Great Expectations), Schema Design, Vector Databases (FAISS, Pinecone)
- **Security & Governance:** JWT + Token Rotation, OAuth 2.0, MFA/TOTP (QR-based 2FA), 5-Role RBAC (71 granular permissions), Audit Logging (365-day / 7-year retention), PII Handling, Encrypted Storage, Transient-Processing Architecture, Human-in-the-Loop Workflows, PIPEDA-Aware Design, OSFI Model Risk Awareness
- **BI & Analytics:** Tableau, Power BI (Advanced DAX), Matplotlib, Plotly, Seaborn, KPI Dashboards, Executive Reporting, Agile/Scrum

DOMAIN EXPERTISE

- **Financial Services & Banking:** Fraud Detection, AML Pattern Recognition, Credit Risk Scoring, Loan Approval Prediction, Customer Analytics & Segmentation, Document Intelligence (OCR), Regulatory Compliance Analytics, Explainable AI for Audits, Anomaly Detection in Transactions
- **AI SaaS & Voice AI:** Real-Time LLM Voice Applications, Agentic Tool Systems, Multi-Tenant SaaS Architecture, Billing & Subscription Systems, RBAC & Enterprise Security, Multi-Language Voice AI (9 languages), Telephony Integration (Twilio)
- **Telecom & Agriculture:** ML Model Migration (R → Azure ML), Animal Behavioral Analytics, Time Series Activity Data, Azure ML Operationalization, Statistical Modeling for Predictive Analytics
- **Healthcare (Design-Ready):** Clinical Document Intelligence, PHIPA-Aware Architecture, Patient Triage AI, Discharge Summary Parsing, PII Redaction Pipelines, Confidence-Gated Human Review Workflows

PROFESSIONAL EXPERIENCE

Lead AI/ML Engineer — Production AI SaaS | Jay Analytix Inc (DeskIQ.ai)

Aug 2025 – Present

Windsor, ON · Remote · GCP · Twilio · GPT-4.1 · Claude Sonnet 4.5

Sole backend engineer for DeskIQ.ai — production AI Voice Receptionist SaaS. Built entire backend from scratch: ~97K lines Python, 557 FastAPI endpoints, 42 modules, deployed on GCP. Telephony via Twilio; LLM inference via GPT-4.1 and Claude Sonnet 4.5.

- **Agentic AI System:** Designed agentic architecture with 8 custom tool schemas (booking, availability, reschedule, customer lookup) + FastAPI execution endpoints. LLM autonomously selects and invokes tools mid-conversation using JSON schema descriptions and system prompt engineering. Zero-shot intent classification gate routes callers to KB retrieval vs tool execution — eliminating unnecessary API calls.
- **LLM Applications & RAG:** Built knowledge base management layer (full CRUD lifecycle), retrieval configuration tuning (top_k, similarity threshold), hallucination prevention via negative example anchoring (🚫 markers in system prompt), and KB-conditional routing in code. Designed dual-prompt architecture: 462-line inbound system prompt / 19-line outbound batch prompt — 95% token reduction on batch campaigns.
- **Production ML Analytics:** Post-call analytics pipeline: webhook handler → MongoDB aggregation → dashboard APIs. Composite success metric (answered + duration > 30s + sentiment). Complexity scoring engine for LLM cost/latency prediction. p50/p95/p99 latency monitoring, per-call cost tracking, tool call success rate dashboards.
- **Security & Responsible AI:** JWT + token rotation, Google OAuth 2.0 SSO, MFA/TOTP (QR-based 2FA), 71-permission RBAC (5 roles) enforced via require_permission() middleware on every route, 365-day audit logging with 7-year retention, human-in-the-loop multi-step confirmation gates for high-stakes AI actions, idempotent Stripe webhook processing.
- **MLOps & Infrastructure:** GKE with HPA autoscaling (min 2, max 20 replicas), GitHub Actions CI/CD (lint → pytest → Docker build with git SHA tag → Cloud Run deploy → health check gate), MLflow model versioning, Prometheus + Grafana monitoring. Resolved production latency incident: concurrency limiting + dual-prompt cut p99 from 8s to 1.2s.
- **NLP & Structured Extraction:** Spoken-email parser (14+ phonetic/regional variants → validated email), datetime parser (40+ spoken-time formats → ISO-8601 + timezone normalization), multilingual voice AI supporting 9 languages with Devanagari-to-Latin transliteration. These extraction patterns directly applicable to document/resume parsing.

AI/ML Engineer (Intern) | Jay Analytix Inc

Feb 2025 – Aug 2025

Ontario, Canada · Remote · Azure SQL + GCP

- **Document-Aware RAG Chatbot:** Built RAG-based chatbot (intent understanding → semantic search → dynamic SQL generation → Matplotlib/Plotly visualization) — accelerating reporting by 40%.
- **NLP Extraction Pipeline:** Developed structured extraction pipeline for semi-structured ServiceNow exports — handling inconsistent schemas, null fields, and format variations across 50K+ records.

Lead Data Scientist — Fraud Detection, OCR & Risk Analytics | Dr. Reddy's Foundation (Client: Vreedhi Financial Services)

Jun 2023 – Jan 2025

Hyderabad, India · On-site · Regulated Fintech · Azure ML · R · Python

Led end-to-end ML delivery for regulated lending fintech — problem framing through Azure ML production deployment, monitoring, and stakeholder reporting. Partnered directly with risk, compliance, finance, and audit teams.

- **Fraud Detection — 90% Precision (~\$500K savings):** Computer vision models (YOLOv3, Faster R-CNN, TensorFlow) detecting document tampering in loan applications — forged IDs, fake bank statements, manipulated income proof. 90% precision after threshold tuning with risk team. SHAP explainability for compliance audit documentation. Deployed to Azure ML GPU compute cluster for training, CPU endpoint for inference.
- **OCR Document Intelligence — 50% Manual Reduction:** Azure AI Document Intelligence (Form Recognizer) pipeline for high-volume regulated financial documents — income statements, bank statements, payslips, identity documents. Table extraction, key-value parsing, structured JSON output to downstream loan systems. Validation rules against regulatory minimums. PII-aware: encrypted storage, access-controlled outputs, automatic redaction before logging.
- **XGBoost Loan Approval Model:** End-to-end pipeline: preprocessing → feature engineering → grid search hyperparameter tuning → 5-fold stratified CV → ~70% accuracy. SHAP interpretability: per-prediction feature contributions for risk team. Deployed to Azure ML managed online endpoint with scoring script (init() + run()). Azure DevOps CI/CD with evaluation gate (precision ≥ 90% required for production promotion).

- **R Statistical Modeling:** Built predictive and segmentation models in R — regression models, time series forecasting, and analytical pipelines using caret, ggplot2, dplyr, data.table. Prototyped models in R for rapid statistical exploration, ported finalized models to Python for Azure ML production deployment. R → Azure ML migration pattern directly applicable.
- **Azure ML MLOps:** MLflow experiment tracking (hyperparameters, metrics, artifacts per run), Azure ML Model Registry with version tags, Azure DevOps CI/CD pipeline with evaluation gates, Azure Application Insights for endpoint monitoring, Azure ML data drift monitor (PSI-based, automated retraining trigger when PSI > 0.2).

Data Scientist — Predictive Analytics & Segmentation | Dr. Reddy's Foundation (Client: Vreedhi Financial Services)

May 2021 – May 2023

Hyderabad, India · On-site · Azure ML · R · Python

- **Customer Segmentation (~20% uplift):** Extra Trees segmentation engine (82% accuracy) across 100K+ customer records. ML-driven targeting drove ~20% campaign engagement increase.
- **Credit Risk / Loan Approval:** XGBoost loan approval pipeline with SHAP interpretability deployed to Azure ML endpoint.
- **Regulatory Reporting:** Power BI + Tableau dashboards translating complex model outputs into compliance-ready visualizations for audit and leadership teams.

Data Analyst — Python, R, SQL & Business Analytics | Vinfosoft Solutions PVT LTDDec 2018 – Apr 2021

Hyderabad, India · On-site

- **R Analytics:** Demand forecasting (ARIMA, seasonality decomposition, regression) and ggplot2 visualization for executive reporting. R-based statistical pipelines for sales and operations datasets.
- **Python + SQL:** Automated ETL pipelines, 3x SQL query optimization, 40% manual reporting reduction. Forecasting models supporting business planning.

PROJECTS — JAY ANALYTIX RESEARCH PRACTICE

FinRoute AI — Intelligent Financial Document Intelligence with SLM/LLM Routing

Production finance document intelligence pipeline built at Jay Analytix. Routes tasks intelligently between SLM (Phi-3 Mini) and LLM (Claude Sonnet 4.5) based on a 4-signal complexity classifier. Simple tasks (field extraction, Q&A, summaries) → SLM at <50ms. Complex tasks (fraud detection, anomaly analysis, risk assessment) → LLM. Auto-escalation if SLM confidence < 0.7.

- **Architecture:** 4-signal complexity classifier (keyword detection, entity count, reasoning depth, risk flag), XGBoost fraud detector, Isolation Forest anomaly detection, SHAP explainability, FAISS vector store for RAG Q&A, FastAPI backend (5 endpoints: /analyze, /extract, /detect-fraud, /anomaly, /qa).
- **Impact:** 73% of queries handled by SLM — 8x cost reduction vs full LLM pipeline. Fraud detection at 90%+ precision. Deployed on GCP Cloud Run with MLflow tracking and GitHub Actions CI/CD.
- **Stack:** Python · FastAPI · Phi-3 Mini · Claude Sonnet 4.5 · XGBoost · FAISS · Azure AI Document Intelligence · GCP Cloud Run · MLflow · Prometheus

EDUCATION

Post-Graduate Diploma — Data Analytics in Business

St. Clair College, Windsor, ON | GPA: 3.25/4

Jan 2025 – Apr 2026

Post-Graduate Program in Data Science

INSOFE, India | Grade: A | National Hackathon: Ranked 8 / 120 Teams

Mar 2020 – Mar 2021

Bachelor of Engineering — Electronics & Communications

SRIT Engineering College, India

Jan 2013 – Jul 2017

CERTIFICATIONS & AWARDS

- Microsoft Certified: Azure Data Scientist Associate (DP-100) — Microsoft
- Prompt Engineering & Programming with OpenAI — Columbia+ (2026)
- Introduction to Responsible AI — Google Cloud (2025)
- Introduction to Large Language Models — Google / Coursera
- Fundamentals of Generative AI & Computer Vision — Microsoft Learn
- Annual Best Performance Award (2022) · Ace of Initiative Compliment (2021) — Dr. Reddy's Foundation
- National Hackathon: Ranked 8 / 120 Teams — INSOFE Data Science Program